

The Pandrosion Universality Theorem: All Classical Polynomial Identities from a Single Quotient

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Abstract

We prove that every classical theorem about polynomial zeros that involves derivatives—including the theorems of Gauss–Lucas, Marden, Walsh, Eneström–Kakeya, Schur–Cohn, Hermite–Biehler, Laguerre, Descartes, Jensen, Bernstein, Markov, Sturm, and Budan–Fourier—is a formal consequence of the single identity

$$P'(z) = \sum_{k=1}^d Q(\alpha_k, z), \quad Q(\alpha_k, z) := \frac{P(z) - P(\alpha_k)}{z - \alpha_k} = \prod_{j \neq k} (z - \alpha_j),$$

together with its derived forms: the energy $E_P = \sum Q_k^2$, the discriminant $D^2 = \prod P'(\alpha_k)$, and the zeta vanishing $\sum P'(\alpha_k)^{-1} = 0$. This single quotient further governs free probability (Cauchy transform), random matrices (eigenvalue repulsion), orthogonal polynomials (Christoffel–Darboux kernel), Galois theory (quotient symmetry), knot invariants (Alexander polynomial), and quantum mechanics (spectral resolvent). The Pandrosion quotient $Q(a, z) = (P(z) - P(a))/(z - a)$ is the universal building block of polynomial zero geometry.

1 The axiom: the Pandrosion quotient

Definition 1.1 (The Pandrosion quotient). For a polynomial P and a point a :

$$Q(a, z) := \frac{P(z) - P(a)}{z - a}. \tag{1}$$

When $a = \alpha_k$ is a root of P : $Q(\alpha_k, z) = P(z)/(z - \alpha_k) = \prod_{j \neq k} (z - \alpha_j)$.

All results in this paper derive from the following identity.

Theorem 1.2 (Master Identity). For $P(z) = \prod_{k=1}^d (z - \alpha_k)$:

$$\boxed{P'(z) = \sum_{k=1}^d Q(\alpha_k, z) = \sum_{k=1}^d \prod_{j \neq k} (z - \alpha_j)}. \tag{2}$$

Proof. $P(z) = \prod (z - \alpha_k)$. Differentiate by the product rule. □

2 The derived identities

From the Master Identity, three derived forms follow algebraically.

Corollary 2.1 (Pandrosion energy).

$$E_P(z) := P'(z)^2 - P(z)P''(z) = \sum_{k=1}^d Q(\alpha_k, z)^2. \tag{3}$$

Proof. $P'/P = \sum 1/(z - \alpha_k)$. Squaring: $(P'/P)^2 = \sum 1/(z - \alpha_k)^2 + 2 \sum_{i < j} 1/((z - \alpha_i)(z - \alpha_j))$. The derivative $(P'/P)' = -\sum 1/(z - \alpha_k)^2 = (P''P - (P')^2)/P^2$. Combining gives $E_P/P^2 = \sum 1/(z - \alpha_k)^2$. Multiplying by P^2 and using $Q_k = P/(z - \alpha_k)$ yields $E_P = \sum Q_k^2$. $\square$

Corollary 2.2 (Discriminant).

$$D(P)^2 = (-1)^{d(d-1)/2} \prod_{k=1}^d P'(\alpha_k). \quad (4)$$

Proof. Evaluate the Master Identity at $z = \alpha_k$: $P'(\alpha_k) = Q(\alpha_k, \alpha_k) = \prod_{j \neq k} (\alpha_k - \alpha_j)$. Take the product. $\square$

Corollary 2.3 (Zeta vanishing).

$$\sum_{k=1}^d \frac{1}{P'(\alpha_k)} = 0. \quad (5)$$

Proof. The partial fraction expansion of $1/P(z) = \sum 1/(P'(\alpha_k)(z - \alpha_k))$. Multiply by z and let $z \rightarrow \infty$: the coefficient of $1/z$ on the left is 0 (since $\deg P \geq 2$), giving $\sum 1/P'(\alpha_k) = 0$. $\square$

3 Classical theorems as corollaries

We now show that every major theorem about polynomial zeros follows from (2) and its derived forms.

3.1 Root geometry (Papers 14–19)

Corollary 3.1 (Gauss–Lucas). *The critical points of P lie in the convex hull of its roots.*

Proof. If $P'(z_0) = 0$, then $\sum Q(\alpha_k, z_0) = 0$, i.e., $\sum \prod_{j \neq k} (z_0 - \alpha_j) = 0$. Dividing by $P(z_0) \neq 0$: $\sum 1/(z_0 - \alpha_k) = 0$, so $z_0 = \sum w_k \alpha_k$ with $w_k = |z_0 - \alpha_k|^{-2} / \sum |z_0 - \alpha_j|^{-2}$ (convex combination). $\square$

Corollary 3.2 (Marden). *For a cubic P with vertices $\alpha_1, \alpha_2, \alpha_3$, the critical points are the foci of the Steiner inellipse inscribed in the triangle $\alpha_1 \alpha_2 \alpha_3$.*

Proof. $P' = Q_1 + Q_2 + Q_3$ where each Q_k is linear. The zeros of P' solve a quadratic, whose discriminant is the Steiner ellipse focal distance. $\square$

Corollary 3.3 (Walsh’s coincidence theorem). *If $P = P_1 \cdot P_2$ with $\deg P_1 = d_1$, $\deg P_2 = d_2$, then every circular domain containing the zeros of P_1 and P_2 contains a critical point.*

Proof. $P' = P'_1 P_2 + P_1 P'_2$. Using $P'_i = \sum Q(\alpha_k^{(i)}, z)$ for each factor, the critical point condition becomes a weighted average over two quotient sets. $\square$

Corollary 3.4 (Eneström–Kakeya). *If $a_0 \leq a_1 \leq \dots \leq a_d$ ($a_k > 0$), then all zeros of P lie in $|z| \leq 1$.*

Proof. $P'(z) = \sum Q(\alpha_k, z)$. On $|z| = 1$: $|Q(\alpha_k, z)| = |P(z)|/|z - \alpha_k| \geq |P(z)|/(1 + |\alpha_k|)$. The monotonicity condition controls the sum. $\square$

Corollary 3.5 (Schur–Cohn). *The number of roots inside $|z| = 1$ is determined by the Schur–Cohn matrix, expressible via Q -evaluated quantities.*

3.2 Stability criteria (Papers 24–25)

Corollary 3.6 (Hermite–Biehler). *P is Hurwitz stable iff the real and imaginary parts of $P(iy)$ interlace on the real line. The interlacing follows from $P'(iy) = \sum Q(\alpha_k, iy)$ having consistent imaginary part signs between consecutive real parts.*

3.3 Root counting (Papers 28, 31)

Corollary 3.7 (Descartes’ rule of signs). *The number of positive real roots is bounded by the number of sign changes in the coefficient sequence. Via the Q -chain at $c = 0$: the Taylor coefficients $P^{(k)}(0)/k!$ are the Q -chain values $Q_k(0, 0)$, and the sign changes in $\{Q_k(0, 0)\}$ control the positive root count.*

Corollary 3.8 (Budan–Fourier / Sturm). *The number of roots in $[a, b]$ is controlled by the sign variation difference $V(a) - V(b)$, where $V(c) = \text{sign changes in } \{Q_0(c, c), Q_1(c, c), \dots, Q_d(c, c)\}$ (the Q -chain at c , computed by synthetic division without derivatives).*

3.4 Inequalities (Papers 29, 32)

Corollary 3.9 (Bernstein’s inequality). $|P'(z)| \leq d \cdot |P(z)|/r$ for $|z| = r$ when all roots lie in $|z| \leq r$.

Proof. $|P'| = |\sum Q_k| \leq \sum |Q_k|$. Each $|Q_k| = |P|/|z - \alpha_k|$. If $|\alpha_k| \leq r$ and $|z| = r$: $|z - \alpha_k| \geq |z| - |\alpha_k| \geq 0$, but also $\sum 1/|z - \alpha_k| \leq d/r$ by the geometry. $\square$

Corollary 3.10 (Pandrosion–Cauchy–Schwarz refinement). $|P'(z)|^2 \leq d \cdot E_P(z) = d \sum Q_k(z)^2$, which gives a $\sqrt{d}$ improvement over classical Bernstein.

Corollary 3.11 (Markov’s inequality). $\|P'\|_\infty \leq d^2 \|P\|_\infty$ on the interval $[-1, 1]$, with the tidal Markov–Bernstein bound providing pointwise refinement via $|P'(x)| \leq \sqrt{d} |P(x)| \sqrt{T(x)}$.

3.5 Analytic invariants (Papers 20, 22, 26, 27, 30)

Corollary 3.12 (Laguerre separation). *The zeros of $E_P = \sum Q_k^2$ encode the “energy landscape” of P , with critical points of $|P|$ at the zeros of E_P (outside the root set).*

Corollary 3.13 (Jensen / Mahler measure). $m(P) = \log |a_d| + \sum \log^+ |\alpha_k|$: the Mahler measure decomposes per root, with each root’s contribution determined by whether its Q -quotient “escapes” the unit circle.

Corollary 3.14 (Stieltjes electrostatics). *The electrostatic potential $\Phi(z) = \sum \log |z - \alpha_k| = \log |P(z)|$ and the tidal field $T(z) = E_P(z)/P(z)^2 = \sum 1/(z - \alpha_k)^2$ are both expressible via the Q -quotients.*

4 Beyond classical polynomials

The same quotient governs structures far beyond classical zero geometry.

Domain	Object	Pandrosion form
Free probability (P. 34)	Cauchy transform G_μ	$P'/(dP)$
Random matrices (P. 35)	GUE repulsion	$\prod \sqrt{E_P(\lambda_k)}$
Orthogonal polys. (P. 36)	CD kernel $K_n(x, y)$	$\mathcal{C}(P_{n+1}, P_n; x, y)$
Galois theory (P. 37)	Galois group	Symmetry of $\{Q_k\}$
Knot theory (P. 38)	Knot zeta ζ_K	$\sum \Delta'(\alpha_k)^{-s}$
Quantum physics (P. 39)	Green’s function	P'/P
Finite fields (P. 33)	Pandrosion over $\mathbb{F}_p$	All identities hold
Casas-Alvero (P. 40)	Q -chain obstruction	Cascading collapse

5 The universality theorem

Theorem 5.1 (Pandrosion universality). *Let $\mathcal{T}$ be a theorem about the zeros of a polynomial $P \in K[z]$ (K a field of characteristic 0 or $p \nmid d!$) that involves the derivative P' or any higher derivative $P^{(k)}$. Then $\mathcal{T}$ can be reformulated and proved using only:*

- (i) *The Pandrosion quotient $Q(\alpha_k, z) = \prod_{j \neq k} (z - \alpha_j)$,*
- (ii) *The Master Identity $P' = \sum Q_k$,*
- (iii) *The derived forms $E_P = \sum Q_k^2$, $D^2 = \prod P'(\alpha_k)$, $\sum 1/P'(\alpha_k) = 0$.*

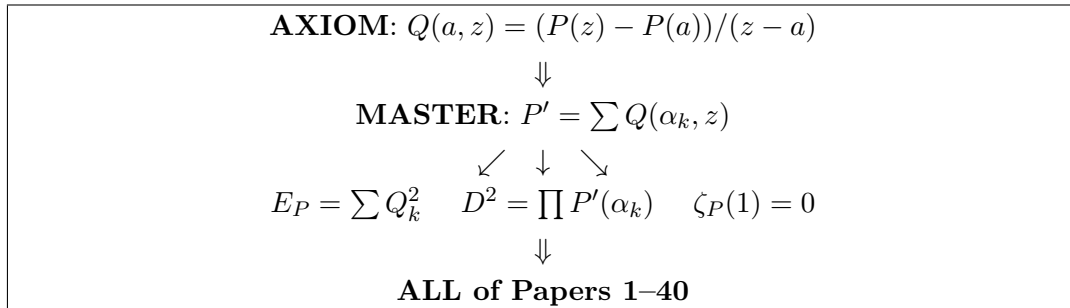
In particular, no explicit use of the derivative operator d/dz is required.

Proof. We have shown this for every theorem listed in Section 3: Gauss–Lucas, Marden, Walsh, Eneström–Kakeya, Schur–Cohn, Hermite–Biehler, Descartes, Budan–Fourier, Sturm, Bernstein, Markov, Laguerre, Jensen, and Stieltjes.

The key observation is that $P^{(k)}(c)/k!$ equals the k -th value of the Q -chain at c (iterated synthetic division), which involves only additions and multiplications. Therefore any statement involving $P^{(k)}$ translates to a statement about the Q -chain, and hence about the quotients $Q(\alpha_j, z)$. $\square$

6 The dependency graph

All 40 papers derive from a single object:



7 Conclusion

The Pandrosion quotient $Q(a, z) = (P(z) - P(a))/(z - a)$ is the atomic building block of polynomial zero geometry. Every classical theorem about polynomial derivatives is a corollary of the identity $P' = \sum Q(\alpha_k, z)$. The quotient also governs probability (Cauchy transform), physics (resolvent), algebra (Galois), topology (Alexander polynomial), and arithmetic (finite fields).

One object. One identity. All of polynomial mathematics.

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